

THE PHYSIOLOGICAL BACKGROUND

Psychology and physiology

If a significant number of the fundamental categories of twentieth-century Psychology had their source in empiricist mental philosophy, could one say that this Psychology was a kind of continuation of empiricist philosophy by other means? That would be going too far. First of all, these 'other means' – experimental and statistical technologies – were never mere passive instruments, but played a crucial role in reshaping the conceptual landscape of the discipline. Secondly, modern Psychology relied on the older sciences for more than its technology. Some of its most basic categories had a biological origin. There was nothing arbitrary about the title of the first major textbook of the new science: *Principles of Physiological Psychology* (Wundt, 1874). The topics, the methodology and the conceptual apparatus were all to be found in the physiology of the time. Strong traces of this legacy were still evident a century later.

The dependence of early experimental psychology on previous work in physiology has always been obvious. What has been much less obvious is the appropriate time frame for assessing the nature of this dependence. It is clear that virtually all of the work in the first psychological laboratories stood in a direct line of descent from similar work in mid-nineteenth-century physiological laboratories. One thinks of work on vision, touch and hearing, on reaction time measurement, on the recording of physiological reactions, and so on. However, that is not all there is to the story. The conceptual link between physiological and psychological topics goes back rather further than this practical link. In fact, the boundary between physiology and psychology was not always drawn as sharply as it was in the latter part of the nineteenth century. By that time, physiology had finally embraced a consistently mechanistic approach. Henceforth it would limit itself to the investigation of somatic functions in a purely physicalistic framework.¹ That project was extremely successful, but it did entail the jettisoning of that part of its older mandate which had extended to psychological issues. In a manner of speaking, the newly autonomous physiological psychology had been spat out by nineteenth-century physiology.

Prior to these developments, physiological texts had dealt with psychological issues as a matter of course. Johannes Müller's great compendium, *Elements of Physiology* (1838–42), had devoted large sections to topics that were later to be considered as falling on the psychological side of the boundary between the disciplines. The same was true of the text of his predecessor, Rudolphi (1821–23). Burdach, another prominent physiologist of the time, explicitly defined the task of physiology as including the study of psychological aspects (1826: 4). In Central Europe this tradition went back at least to the monumental mid-eighteenth-century text *Elementa Physiologiae* (translated as *First Lines of Physiology*, 1966) by Albrecht von Haller. In Britain, there was a late efflorescence of psycho-physiological texts even beyond the middle of the nineteenth century, prominent examples being Thomas Laycock's *Mind and Brain* (1860), Sir Henry Holland's *Chapters on Mental Physiology* (1852), and William Benjamin Carpenter's *Principles of Mental Physiology* (1874). This last work was republished in America as late as 1891 and was cited extensively by William James.

The changing nature of the relationship between psychology and physiology had profound consequences for psychological language. Could the same set of terms be used for describing both bodily and mental phenomena, or would two different languages be needed? At one extreme there was the consistently dualistic position which regarded with horror any mixing of the physicalistic language of physiology and the mentalistic language of psychology. Nineteenth-century physiological mechanists took this position, but so did many of the moral philosophers who were their contemporaries. Both sides feared contamination by the other. However, ordinary language was not so fastidious in its distinctions and contained many terms that carried a mixed somatic and psychological meaning. Common examples were attitude, motive and temperament. Historically, the separation of physical and mentalistic language was a modern development, and in the nineteenth century it was far from complete. The development of Psychology as an autonomous scientific discipline entailed a confrontation of these issues. Two sets of linguistic boundaries would have to be established: the language of scientific Psychology had to be distinguished from everyday usage, and it had to be found a place somewhere between the non-communicating solitudes of physiology and moral philosophy.

There were two areas of investigation in which the boundary between physiology and psychology was particularly unclear. One was the area of sensation. Studies of the sense organs were certainly physiological, but the activity of these organs involved the property of sensibility which was usually considered to be psychological. The second area with blurred boundaries was that of animate motion. The explanation of the movements of living creatures and living organs had become a primary task of physiology. But these movements often exhibited such marvellous co-ordination and apparent wisdom that the question of

their direction by some kind of mind or soul kept on coming up. In such discussions no sharp line between the physiological and the psychological could be drawn. It was only the victory of physiological mechanism and physicalism, towards the middle of the nineteenth century, that established rigid borders around the subject matter of physiology. Even then, physiological reductionism applied to psychological topics could still flourish.

At different times, Psychology adopted physiological categories that differed widely in their degree of generality, from highly specific, technical categories to broad categories whose use was pervasive. We will focus on the latter. Among them the interconnected categories of organism, stimulus, reflex and energy require special attention because of their importance for the emerging conceptual structure of modern psychology.

The first of these, *organism*, is fundamental for the transfer of physiological concepts into psychology. It establishes a category of beings that unites animals and humans and thus establishes the plausibility of assuming similar attributes for both. If humans are taken to be organisms in some fundamental sense, then the defining features of what it means to be an organism will apply to them. During the major part of the twentieth century, the most visible of these features have been those related to the theory of evolution by natural selection, i.e. features like adaptability, 'survival of the fittest', and heredity. But the concept of the organism is older than Darwinian theory. Before that theory achieved its pre-eminent position, the nature of organisms was defined by other features. Those features have not disappeared. Some have simply become invisible through being taken for granted; the role of others is now seen as more limited. Response to stimulation is an example of the first case, reflex action and energy are examples of the second. But this does not affect the historical importance of these attributes of organisms, for the traces of their former importance are still present.

For example, *stimulation* is probably the most pervasive category in all of twentieth-century Psychology, precisely because it is so deeply embedded in the conceptual and historical foundations of the discipline that it has acquired a completely taken for granted quality. Whatever the specific content of modern psychological descriptions or theories, a broad range of them presupposes that their domain of application has already been formulated in terms of the category of stimulation. Accounts of perceptual processes deal with the effects of present stimulation, accounts of learning and memory with the effects of past stimulation. Motives and emotions are often conceptualized in terms of 'internal' stimulation, and a still strong tradition conceives of all human behaviour as a 'response' to stimulation. A great deal of psychological description is based on a pre-theoretical analysis of complex and potentially ambiguous situations in terms of stimuli and their effects. A great deal of psychological theorizing is designed to explain how such

effects work. The relevance of such theories depends on the prior description of phenomena in terms of the category of stimulation. So pervasive, and so deeply rooted, is the use of this category that its implications have become invisible. It is assumed to be completely neutral and to entail no assumptions about the nature of the subject matter that is to be described or explained.

That, however, is an illusion. Description of phenomena in terms of stimuli and their effects imposes a certain structure on the phenomena. In most instances there are other descriptions that could have been adopted but have been quietly rejected. Moreover, the category of stimulation is a historical product which emerged in specific contexts and for specific reasons. To understand its significance we need to appreciate the circumstances of its historical formation and its subsequent trajectory. Stimulation has not always meant the same thing. In the course of its history, the connotations of the term have varied considerably. At each stage, a different aura of meaning was imposed when phenomena were subsumed under the category of stimulation. At every stage, however, this term, perhaps more than any other, expressed fundamental convictions about the status occupied by subjectivity in the natural order.

The present chapter explores the 'archaeology' of stimulation by exposing some of the deeper conceptual strata that have provided the foundations on which modern variants could be erected. Three layers are considered here: the deepest is vitalistic, the next involves a transition to mechanistic models in the shape of the concept of the reflex, and the third represents the incursion of metaphorical physicalism through nineteenth-century concepts of energy.

Vitalistic roots of 'stimulation'

Both the area of sensation and the area of animate motion were deeply affected by a fundamental reformulation of physiological problems that took place in the second half of the eighteenth century. The concept of stimulation played a crucial role in this reformulation. In one sense, what happened was less a question of the reformulation of physiological problems than a question of reformulating problems *as* physiological. In the late seventeenth and early eighteenth centuries the important work on the operations of the living body had been conducted within a dualistic framework. Descartes had set the tone. Souls and bodies represented entirely different orders of being. Living bodies were pieces of physical machinery whose reactions did not require any explanation beyond those supplied by the science of mechanics. The living structures that conveyed the effects of stimulation were regarded as mechanical contrivances that behaved like strings, valves, tubes, and fluids under pressure. The problem of the production of animate motion was not

distinguished in principle from the problem of the production of inanimate motion. Physiological problems were problems in applied mechanics. At this time, the term 'stimulus' is occasionally found in the medical literature (Möller, 1975) with reference to external agents that produce injuries (e.g. burns and stings) or that alleviate pathological conditions (e.g. the reduction of swellings).

The emergence of the category of stimulation as a fundamental element in the construction of uniquely *physiological* questions is intimately tied up with the switch from mechanism to vitalism that became increasingly pronounced in the latter part of the eighteenth century. Vitalism came in many varieties (Benton, 1974), but here we are only concerned with their common denominator, namely, an insistence that the processes of life could not be reduced to physical mechanism but involved the operation of special principles characteristic of living forms. The much respected Haller, for example, considered that *sensibility* was due to a unique property of living nerves, and that animate motion depended on the special *irritability* of muscles (Haller, 1922). With the advance of physical and chemical knowledge such principles seemed, at best, redundant, and at worst, unscientific and mystifying. However, the bad press which vitalism got from the new breed of mechanists in the nineteenth century should not blind us to the fact that some of the conceptual constructions of the vitalists became part of the common language of physiology. Among these constructions the notion of stimulation played a crucial role.

In the old dualistic scheme, only two kinds of instigators of activity were recognized. These were physical forces and acts of will. The activity of animals was entirely due to the former, while human activity was the result of both. Cartesianism held that the passions were produced by physical events in the body, but they could be opposed by the soul's rational will. This scheme created an unbridgeable gulf between the domain of natural science, concerned with animate and inanimate bodies, and the domain of the soul, left to the tender mercies of theologians and philosophers. The vitalistic physiology of the eighteenth century began to bridge this gulf by constructing a third kind of activity which was neither like the mechanical action of inanimate bodies nor the deliberate rationality of human moral action (Danziger, 1983). It was activity in response to a stimulus, and it was characteristic of living beings, now thought of as *organisms*. Whereas inanimate motion depended on a transmission of physical force through passive structures, so that variations in the effect were entirely accounted for by variations in the impinging cause, animate motion depended on the relationship between stimulus and living body. This meant that the effects of stimuli could not be predicted merely from a knowledge of physical principles. Weak stimuli could have powerful effects, and out of several mechanically possible effects only those occurred which preserved the integrity of the living system.

The category of 'motion from a stimulus' first occurs in the work of the prominent Scottish physiologist, Robert Whytt (1768). Such motions were governed neither by the laws of mechanics nor by the laws of logic, but obeyed their own kind of 'necessity' which had to do with the preservation of organismic integrity. Whether an instigator of vital movement was physical or mental had nothing to do with its status as a stimulus:

The remembrance or idea of substances, formerly applied to different parts of the body, produces almost the same effect as if those substances themselves were really present. Thus the sight, or even the recalled idea of grateful food causes the saliva to flow into the mouth of a hungry person; and the seeing a lemon cut produces the same effect in many people. The sight of an emetic, nay its very name, or seeing others vomit, will in many delicate persons raise a nausea. (Whytt, 1768: 133)

Responses to stimuli were involuntary, but their causal determination was not a mechanical one. There was recognition of a new category of causally determined activity, where the causality was non-mechanical and the mental or physical nature of the cause was irrelevant. That shattered the old dualism of involuntary and voluntary action, the former having physical causes, the latter a mental cause, the will. It also changed the conception of what mind was:

The mind, therefore, in carrying on the vital and other involuntary motions, does not act as a rational, but as a sentient principle; which, without reasoning, is as certainly determined by an ungrateful sensation or *stimulus* affecting the organs, to exert its power, in bringing about these motions, as is a scale which by mechanical laws turns with the greatest weight. (Whytt, 1768: 152)

Whytt still retained the concept of a rational soul capable of rational consciousness and volition. But he added the construct of a 'sentient soul' which lacked freedom, and whose rationality was not that of self-reflective reasoning but that of the preservation of the biological organism.²

One important consequence of this was that voluntary and automatic actions were no longer seen as belonging to irreconcilably separate categories but that voluntariness and automatism were matters of degree. A particular activity might involve varying degrees of automatism at different times and under different conditions. This was a considerable breach in the conceptual fortress that had protected the realm of the will from any kind of naturalistic encroachment. As we saw in the last chapter, the philosophy of Hume had worked in the same direction, and even more radically. However, Hume's solution took the form of a mechanization of the mind, while retaining the separation of mind and body. Compared to this, the new category of stimulated motion was much more productive. It opened up a space for concrete scientific investigation that undermined the traditional position more subtly and more effectively than a purely philosophical critique.

In the work of William Cullen (1710–90), Whytt's successor at Edinburgh, the possibilities of the stimulus concept were developed further. Where Whytt had mainly been concerned with the role of stimuli in the internal functioning of the body, Cullen stressed the crucial role of external stimuli in maintaining normal physiological activity (Thomson, 1832). Other medical authors extended the category of stimulated action in a more psychological direction. At the end of the eighteenth century, Erasmus Darwin (Charles Darwin's grandfather) produced a comprehensive theory of human behaviour and experience in terms of three fundamental categories: stimulation, muscular contraction, and a central 'sensorial power' (1794–96). Unlike his medical predecessors, he did not limit his stimulus–response analysis to actions like sneezing, swallowing and vomiting, but extended it to everyday voluntary activity. He was able to do this by drawing on the principle of association developed within empiricist philosophy. This fusion of the physiological schema of stimulated motion with philosophical empiricism had considerable potential, but Darwin was ahead of his time. It was at least half a century before such ideas resurfaced in Britain, and in the meantime Darwin had been forgotten by his countrymen. However, he fared much better on the continent of Europe, especially in Germany. Ironically, some of his more original notions arrived back in his own country in the form given to them in Johannes Müller's textbook, already referred to. (Young, 1970).

The attention paid to Erasmus Darwin in Germany took place against the background of a fast developing interest in the stimulus concept in that country. In fact, after the original contributions of Scottish medical men, the further elaboration of the category of stimulation was largely a German affair. Cartesian dualism had never become as solidly established in Germany as in France and Britain, so that there was much less resistance to non-dualistic formulations in German medical discourse (Verwey, 1985). Leibnizian philosophy, with its emphasis on the principle of continuity in nature – the idea that natural differences in kind were traceable to many much smaller differences – was also favourable to the extension of the schema of stimulated motion.

When it was extended from the reactions of specific biological organs to the reactions of the organism as a whole, the concept of action from a stimulus became psychologically significant. Eventually, it was applied to purely psychological causes and played an important role in the development of a naturalistic basis for psychology. Thus, J.G. Herder, a prominent figure in the German Enlightenment, advocated a psychology tied to physiology 'at every step' and made copious use of the stimulus concept in a somewhat muddled attempt at showing the way (Herder, 1778).

Soon there was an enormous expansion of the field of meanings covered by 'stimulus' (*Reiz*) in the German literature. The term 'internal stimulus' was now used to describe the causal effect of mental activity.

In the writings of philosophically inclined biologists like Treviranus (1822) the will is said to be the stimulus for bodily movements, an action that is said to be analogous to that of external stimuli and to be subject to natural laws. Stimuli are also credited with long term effects through the changes they produce in bodily organs, including the nervous system.

By the end of the eighteenth century, J.C. Reil, a major contributor to the systematization of medical knowledge, was able to distinguish between various senses of the term 'stimulus' (Reil, 1910). In line with Haller, there was the distinction between sensory and motor stimuli, but there were also other distinctions. Among these, two are worth mentioning. First, there was a distinction to be made between external, physical, stimuli, and those stimuli that acted within the organism. The latter need not be physical and occurred in connection with such phenomena as the association of ideas. Secondly, there was a distinction between stimuli in general and those stimuli that corresponded to the specific sensitivity (*Reizbarkeit*) of a particular organ, light for the eye and sound for the ear providing obvious examples. The idea of specific sensitivity was taken up by a later generation of physiologists, like Johannes Müller (1826), and for a time it provided a theoretical framework for empirical work on sensation and even space perception. In its later incarnation it came to be known as the 'doctrine of the specific energy of nerves' (Boring, 1942).

Ramifications of the reflex concept

In its early history, the category of stimulation had acquired a rather broad meaning derived from a vitalistic understanding of causal processes in living organisms. After about 1830, however, vitalism was in decline, experimental techniques were becoming more precise, and scientific explanation became more closely identified with the search for elements. These changes manifested themselves both in the investigation of sensory physiology and in the investigation of animal movement and led to profound changes in the practical deployment of the concept of stimulation. Here we will focus on developments in the area of animal movement because, in the long run, these had the most far-ranging repercussions for the history of psychology.

During the last two-thirds of the nineteenth century, and into the early twentieth century, the scientific study of animal movement was dominated by the concept of the reflex. Stimulation was of course an integral component of this concept, but its meaning had undergone a profound change. Attention had shifted from the broad features of biological causation to more specific effects. It could now be taken for granted that the effects of external agents depended on the organization of the living system, and this recognition was expressed in the retention of the term

'stimulus' in physiology. But questions about the overall integration and functioning of the organismic unit now took second place to questions about the *elements* of the living organization. That implied a change in the way in which this organization was conceived. Increasingly, in the early nineteenth century, the organism was understood in terms of separate functional elements, each with its own spatial localization in the body. Physiological investigation became dedicated to the isolation and identification of such elements.

This emerges very clearly in the work of Marshall Hall on muscular responses that depended on stimulation of the spinal nerves. His experiments on spinal animals were not radically different from earlier work, but his conceptualization certainly was. He attributed the responses of these preparations to the activity of an 'excito-motory', or 'true spinal' system of nerves, which was to be sharply distinguished from the 'sentient and voluntary' part of the nervous system located in the cerebrum. Whereas the reactions mediated by the latter involved conscious sensation and were often 'spontaneous', in the excito-motory system 'there is always the application of an appropriate stimulus, or cause of excitement or irritation; this is followed by the contraction of peculiar sets of muscles' (Hall, 1836: 21). Hall referred to this as 'the reflex function of the spinal marrow', a function composed of many specific reflexes. One could now speak of a reflex in the singular. Reflexes also acquired an anatomical localization in specific nervous pathways. Soon, the concept of the *reflex arc* would become popular. That meant a hard-wired connection from a particular sense organ via sensory nerve, specific spinal segment, and motor nerve to a specific set of muscles.

The decomposition of the reflex function into separate reflex elements led to the replacement of the general function of sensitivity to stimuli (sensibility) by an aggregate of specific stimulus-response elements. It was now assumed that stimulation achieved its effects through potentially identifiable *elementary* units. However complex stimulating situations and their associated actions might be, they were ultimately composed of sensory and motor elements.

These changes were part of a more general transformation in the way of regarding the living organism. Whereas vitalistic physiology had tended to look at specific functions in terms of their contribution to the overall integrity and purpose of the living system, the new physiology was fascinated by the mechanisms at work in the sub-systems. It now seemed more rewarding to trace the material basis of such mechanisms than to speculate about how everything fitted together. The organism was now an assembly of 'parts and pieces in movement' (Figlio, 1977). Such changes went hand in hand with changes in investigative practice. Vitalists like those we discussed in the previous section had certainly conducted experiments, but their experimentation was intended to answer broad questions, often by embedding it in a great deal of purely

observational material. In the nineteenth century, physiological experimentation became increasingly dedicated to answering highly specific questions. At the same time, the experiment became, first, the method of choice, and then the only really scientific method for the expansion of knowledge. An attitude of bold interventionism had replaced the older sense of awe before the marvels of life.

In Hall's version, reflex action explicitly runs off without conscious sensation playing any role. The reflex function is now localized in one part of the nervous system and consciousness in another. Stimulation was no longer regarded as achieving its effects through its link to sensibility but through its link to motor response. The connection between external influence and physical motion had become primary, and the role of mind or consciousness in mediating this connection had been downgraded.

This involved a kind of dualism which had been in abeyance during the period of vitalism. But the issue was not settled overnight. Johannes Müller, a vitalist who studied reflex action at the same time as Hall, always maintained that reflexes were mediated by some form of sensation. Two camps quickly formed around this question, both in Britain and in Germany. On the one side were the dualists, like Hall and his chief ally, R.D. Grainger, who advocated a sharp distinction between *stimulation*, conceived purely physically, and *sensation*, involving consciousness. On the other side were those who clung to an older conception of stimulation that necessarily included some form of sensation.³

One very important consequence of this controversy was a clarification of the meaning of *sensation*. In the late eighteenth and early nineteenth centuries this term had often referred to any kind of impression on a living organism, so that there was no sharp distinction between sensation and stimulation. In fact, the term 'impression' was often used to cover both. In Germany, the reality of 'unconscious sensations', derived from Leibnizian *petites perceptions*,⁴ was widely accepted. Essentially Aristotelian notions had not been completely replaced by Cartesian dualism, so that individual consciousness was *not* regarded as the defining criterion of mind. Mid-nineteenth-century physiological mechanism forced the issue by insisting on a physicalistic language in which 'sensation' had no place.⁵ Stimulation and sensation were decisively separated, the former reacquiring a purely physical meaning, and the latter being limited to conscious events. It should be noted, however, that this distinction reflected the interests of idealist philosophy as much as those of mechanistic physiology (see note 3). It was a very Victorian solution which enabled the physiologists to get on with their work on the nervous system without appearing to threaten a moral order based on the presupposition that individuals were free to make conscious choices.

For the relationship of physiology and psychology these changes were a source of both estrangement and opportunity: estrangement because

there was no longer any place for psychological concepts in the new physiology, and opportunity because a new psychology might be constructed with the help of an invigorated physiology. In the long run, what proved most useful for the establishment of Psychology as a discipline was not so much the new physiological knowledge as the advances in experimental technology that had produced this knowledge. However, this did not become apparent until the last quarter of the nineteenth century, when the new psychological laboratories were established. Before that, there were several determined attempts at establishing a new kind of psychology by using some of the new physiological *concepts* to reformulate psychological issues. Among these concepts, by far the most important was that of the reflex. For nearly a century, from about 1850, it exerted a powerful influence on the conceptualization of psychological issues.

From the beginning, there was impatience with Hall's dualistic limitations on the scope of the reflex concept. The first to claim vast territories for this concept was Thomas Laycock, a mid-nineteenth-century medical authority based at Edinburgh. Rather unusually, he had acquired a definite interest in German philosophy of nature during his early studies at Göttingen. Applying the Leibnizian principle of continuity, he was convinced that the mode of action of the nervous system could not vary fundamentally at its different levels. If reflexes constituted the principle of nervous functioning at the spinal level, as had recently been established, then the cerebrum too must operate on the basis of reflexes (Laycock, 1845). This meant that Hall's rather limited concept of reflex function had been transformed into a general principle governing the entire behaviour of advanced animal organisms. Laycock's version of this idea was only the first of many.⁶

His student, Hughlings Jackson, became a much more influential neurologist than Laycock himself and established the usefulness of the extended reflex concept for clinical practice. He analysed disorders like aphasia in sensori-motor terms, and proposed that, in general, the mental symptoms of cerebral disease were 'due to lack, or to disorderly development, of sensori-motor processes' (Jackson, 1931). Similar views were held by other major figures in late nineteenth-century neurology (e.g. Ferrier, 1876). The appeal of the generalized reflex scheme was not limited to Britain. In 1863 the Russian physiologist, I.M. Sechenov, published a monograph entitled *Reflexes of the Brain* (1965), in which he attempted to analyse a range of higher order functions in terms of the reflex schema. His ideas remained largely unknown outside Russia, but they had an enormous influence on his student, I.P. Pavlov, who became the central figure in the early twentieth-century popularization of the reflex concept. Among Central European physiologists, the generalized conception of the reflex had also become firmly established in the late nineteenth century. Freud's early teachers, Brücke, Meynert and Exner, were all committed to this conception (see Ammacher, 1965; Ellenberger,

1970), and its effects are easily detectable in Freud's own early foray into speculative psycho-physiology (1954).

For early twentieth-century American psychology, it was Sir Charles Sherrington's work that provided a major source of authority in using the reflex as a general category of behaviour. Sherrington had successfully combined the elementaristic with the generalized use of the reflex concept in his experimental studies. He had shown how specific reflexes combined to produce 'integrated' responses of the organism as a whole, particularly the maintenance of its posture. Moreover, he had gone so far as to assign a psychological meaning, 'conative feeling', to reflex reactions guided by distance receptors which involved 'the musculature of the animal as a *whole* – as a single machine' (Sherrington, 1961: 326; original italics). The way was now clear for the use of the reflex as a general *psychological* category applicable to the behaviour of all organisms, including humans. We will look at this development in Chapters 6 and 7.

The metaphor of mental energy

The generalization of the reflex schema favoured the use of physicalistic metaphors for describing and explaining the workings of the mind. If an essentially physical reflex function was responsible for all the actions of higher organisms, then their mental attributes must also be dependent on this function. That dependence became intuitively understandable when one described mental and physical processes in the same, essentially physical, terms. Concepts of *energy* were particularly well suited for this purpose, because, although they had become thoroughly physicalized, one could still talk of mental energy without raising too many eyebrows. The principle of the conservation of energy, and the associated transformation of different forms of physical energy into each other had become established in physics by 1850. After that, talk of mental energy made it possible to invoke the analogy of the transformation of heat into mechanical energy when trying to explain the influence of bodily on mental events, and vice versa. An early suggestion along these lines was made by W.B. Carpenter (1857), a prominent contributor to the discussions that developed around the reflex concept.

Just before mid-century, the Berlin physiologist, Emil Du Bois-Reymond, had provided the definitive empirical demonstration of the electrical nature of the nervous impulse. This had fully legitimized the use of energetic language when speaking of the brain as a physical organ. Reflex processes were now described in terms of nervous 'excitation' and 'inhibition'. Excitation was energy that could be conducted, accumulated and discharged. Late nineteenth-century models of reflex action all made use of this kind of language, and there was a strong temptation to speak of the operations of the mind in the same terms.⁷

One can see this very clearly in Freud's *Project for a Scientific Psychology* (1954). Its central concept was that of a waxing and waning supply of energy or excitation 'occupying' specific neural loci and then being discharged in motor activity. Stimulation, external or internal, is now understood as that which causes the build-up of central excitation. By this time the metaphorical extension of energetic language to mental life had become common, and Freud was no exception. Energetics reappears in psychological form, not only in Freud's 'instinct' theory, but in categories that are used descriptively in psychoanalysis, like 'cathexis'.⁸

The traces of mid-nineteenth-century energeticism within psychological usage were as strong in Britain as in Germany. The language of both major figures of British psychology in that period, Herbert Spencer and Alexander Bain, reflects underlying energetic assumptions. Here we will focus on Bain, whose categorization of psychological phenomena was intimately related to his energeticism. It was through Bain's authoritative texts that this usage became widely accepted in academic psychology.

For Bain, the animal organism is an energy-transforming system which turns the energy derived from food into 'vital energy' that flows into the muscles and is expended in the movements of the organism. The nervous system is no longer a passive conductor of the effects of stimulation but a storehouse of energy that seeks an outlet. Bain accepts 'the physiological fact of a central discharge of nervous energy where no stimulus from without is present as a cause'. Together with other observations, this leads 'irresistibly to the conclusion, that there is in the constitution a store of nervous energy, accumulated during the nutrition and repose of the system, and proceeding into action with, or without, the application of outward stimulants' (Bain, 1977: 328). There is 'a central fire that needs no stirring from without' (1977: 329).

But this is a fire that is not simply physical. A 'coincidence' of states of bodily agitation and states of mental excitement is assumed to exist. Variations in the one produce equivalent variations in the other. More particularly, Bain considers that a rise in vital energy manifests itself as pleasure, while a fall is experienced as pain. These feelings (pleasure, pain, and hedonically neutral excitement), are the basic elements in *emotion*. They define what an emotion is. But because these feelings are manifestations of fluctuations in vital energy, emotion has now become a psycho-physiological category. As we saw in the previous chapter, the category of emotion arose within mental philosophy. For Hume, Kames and Brown 'emotion' described a mental state. Now, however, the mental states so described are defined as dependent on specific physiological events. Emotion is as much physical as it is mental. This has two important consequences for the way this topic is pursued in the late nineteenth and most of the twentieth centuries. On the one hand, talk about emotion is strongly influenced by the energy metaphor, with its images of 'discharge', 'tension', 'diffusion', etc. On the other hand, the

problem of emotion within academic Psychology becomes dominated by psycho-physical questions.⁹

At an even more basic level, the energy metaphor was intimately involved in the emergence of a new distinction between psychological attributes. The energy which psychology borrowed from physiology was purely physical and therefore blind. If organisms were to survive, direction must be imposed on their movements. A separate controlling and directing function was therefore needed to assure the proper channelling of energy. Bain draws a fundamental distinction between 'the directing and the moving powers'. The latter were identified with emotional energy, the former with the will. There was also a third basic power, that of the intellect. But Bain was a good British empiricist, so the intellect's powers were entirely instrumental. This tripartite division of the mind replaced a much older dichotomy between its 'active' and its intellectual powers. The tripartite scheme had originated in the late eighteenth century, when Romantic impulses led to the elevation of feeling as a fundamental category on a par with the will and the intellect. But nineteenth-century energeticism reinterpreted this threefold division. It was no longer the experiential quality of feeling which ultimately defined the third mental power, but its capacity to energize, to *move*.

That, however, entailed a change in the older 'active' power, the will. Earlier on, the will had had the function of translating ideas in the mind into actions in the world. That was no longer necessary, because the physical organism was perfectly capable of motor activity. It had its own energy. What was still needed, however, was something that would give appropriate direction to the discharge of energy in movement. This was the new role of the will. If it was to affect the flow of physical energy, however, its action would have to be thought of in quasi-physical terms. That was accomplished by evoking the image of *channelling*. The will directed action by providing the right channels for energy to flow along. To do this, however, it would have to be assigned a place in the energy economy of the organism. It would have to be regarded as one natural force among others. But in that case, should it still be regarded as a fundamentally distinct category of mental life? Towards the end of the nineteenth century it became increasingly difficult to answer that question in the affirmative, once one had accepted the other popular categories of the current psycho-physiology. The will was in decline, and it would not be long before it disappeared from scientific psychology.

Notes

1. The young German physicalists, Brücke, Du Bois-Reymond, Helmholtz and Ludwig, took the lead in developing an explicit programme along these lines which dominated the development of physiology for the rest of the nineteenth century. In France there had been a more continuous tradition of physiological reductionism. Developments in Britain were slower.

2. This was a great advance on the earlier hypothesis of the German medical theorist, G.E. Stahl, that the operations of the rational soul were involved in the functioning of the body. Stahl was still committed to the dualism of reasoning mind and inert matter.

3. In Germany the controversy culminated in a clash between E. Pflüger, editor of a major physiological journal, and the medically trained philosopher, R.H. Lotze. Pflüger (1853), in a position reminiscent of Whytt a century before, maintained that the adaptive nature of the responses of spinal frogs implied the operation of a 'spinal soul'. Lotze (1853) denied this and drew a sharp line between mechanical responses at the level of spinal reflexes and actions controlled by the conscious soul. Like most dualists, he wanted to preserve an autonomous space for conscious moral decisions. In Britain, a late defender of the Pflüger position was the scientific outsider, George Henry Lewes (1877), who tended to swim against the Victorian stream in other respects as well (see Ashton, 1991).

4. Leibniz had applied his continuity principle to consciousness and perception, making consciousness a matter of degree. Complex perceptions, like the sound of a waterfall, are the product of many minute impressions (*petites perceptions*) that individually are not fully conscious, but that together produce an impression at a high level of consciousness.

5. Another term which used to be indifferent to mind-body dualism, and which was thrown out of physiology at the same time as sensation, is 'sympathy'. For an extended discussion of the controversies which sealed the physiological fate of both terms, see Leys (1990a).

6. See Danziger (1982).

7. The ambiguities of this usage in the case of inhibition have been extensively explored by Roger Smith (1992).

8. The inverted commas around instinct and cathexis are intended as reminders of the fact that these terms are the ones chosen by Freud's translators, not by Freud himself. The terms Freud used were *Trieb* and *Besetzung*. The literal translation of the former is 'drive', of the latter, 'occupation'. Both original terms are more expressive of the underlying energeticism than the English versions. Freud's use of the energy metaphor may have been derived from Gustav Theodor Fechner who used it to justify his invention of psychophysics: 'Just as it takes a certain quantity of kinetic energy to split a log or lift a given weight to a given height, so does it take a certain quantity to think a thought of a given intensity; and energy for one can be changed into energy for the other' (Fechner, 1966: 36).

9. These questions tended to take one of two forms: (a) questions about the recognition of emotions from their physical 'expressive' aspects; and (b) questions about the direction of causal influence between the physical and mental aspects of emotion (the debate about the James-Lange theory).

PUTTING INTELLIGENCE ON THE MAP

Biological roots

In the words of a well-known historian of science: 'Intelligence as currently and conventionally understood by psychologists is a brashly modern notion' (Daston, 1992: 211). Indeed, Baldwin's comprehensive *Dictionary of Philosophy and Psychology*, first published in 1901, does not have a separate entry for 'intelligence'. The reader is simply referred to the term 'intellect', which is defined as 'the faculty or capacity of knowing'. At this point 'intellect' was clearly part of the technical vocabulary of philosophical psychology and was presumed to refer to a distinct psychological entity, but 'intelligence' was a mere synonym without any specific scientific meaning of its own. Two decades later Charles Spearman, the British pioneer of testing intelligence, was still complaining that: 'Right up to the present day a large number – perhaps the majority – of even the best accredited textbooks on psychology do not so much as mention the word "intelligence" from cover to cover' (Spearman, 1923: 2).

Nevertheless, in the early years of the twentieth century 'intelligence' had suddenly become a scientific buzz word, among a significant number of psychologists. British psychologists in 1910 (Myers et al., 1910), and American psychologists in 1921 (Thorndike et al., 1921), organized extensive discussions about the meaning of 'intelligence', never doubting that the term had a distinct reference and paying little attention to 'intellect'.

What had happened? Was this merely a linguistic change, a turn of verbal fashion without any fundamental significance? Or should we regard the verbal change as a sign that more interesting changes were afoot? One such change is obvious: intelligence tests had been invented in the meantime and by 1921 were certainly very much part of the psychological map. But does this tell us why 'intelligence' suddenly acquired a psychological life of its own? In a sense, yes, the existence of a widely popular technology was certainly crucial in creating a distinct referential domain for the newly fashionable term. But then again, no, because have we not merely pushed the question one step back? Why,